

Mahdi Lashkari

Enterprise AI & Data Platform Leader

Portfolio Leadership · Architecture Governance · Agentic AI Systems

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I lead enterprise AI and data platforms from business need through portfolio governance and production systems, including technical judgment on AI investments and on the LLM and agentic systems those programs ship.

Skills

AI Leadership

AI strategy · roadmap · portfolio management · technical leadership · enterprise adoption

Agentic AI & LLM Systems

LangGraph · DSPy · ReAct / tool calling · RAG · context management · knowledge graphs

Architecture & Platforms

DDD · hexagonal architecture · modular monolith · data platforms · AI infrastructure

Engineering Technologies

Python, FastAPI, SQLAlchemy 2, Alembic, PostgreSQL, JWT, SSE, Docker, pytest, GitHub Actions, OpenAI API, AWS Bedrock, NVIDIA H200, PyTorch, LangChain, Django, AWS/GCP, FAISS, Neo4j

Professional Experience

Sina Advanced and Intelligent Technologies Development Co.

Tehran, Iran

Data & Operations Manager — Enterprise AI & Data Platforms

Sep 2025 – Present

- Governed **AI intake** from affiliated organizations: about **48 initiatives** assessed through needs discovery, scoping, RFP and proposal review, architecture and vendor evaluation, and prioritization support.
- Directed **Sina Data**, the national Data Operator program, from stakeholder requirements and platform strategy through architecture, sourcing, staged delivery, and evidence-based acceptance.
- Governed data-to-AI architecture and release readiness: scope held to business value, requirements traced to decisions, and only evidence-backed capabilities treated as ready.
- Participated in technical due diligence of AI investment opportunities, assessing viability, architecture, and implementation risk to support investment decision-making.
- Steered build versus buy, using Snowflake and Databricks, and aligned NVIDIA H200 serving, isolation, and capacity with the AI roadmap.

Chimney CAS (Conversational AI System)

Remote, Part-time

AI Architect & Technical Lead — Agentic AI Systems

Jan 2026 – Present

- Owned Chimney's end-to-end AI architecture and technical direction: intent routing, profile and memory extraction, planning, tool use, evidence-grounded replies, and multi-turn state.
- Architected the orchestration layer with **LangGraph** and **DSPy**: structured routing and extraction, ReAct tool use, provider-agnostic models, runtime configuration, and OpenAI-compatible APIs plus AWS Bedrock.
- Defined the DDD and hexagonal architecture—bounded contexts, public contracts, and **ConversationState**, **TurnPlan**, and **CapabilityOutput** boundaries—so domain policy stays out of orchestration.
- Established trust boundaries that separate LLM proposals from deterministic facts: normalization, resolution, applied semantic deltas, streaming, HTTP/SSE contracts, and architecture regression gates.
- Directed model and tool selection and autonomy limits: deterministic plans and shortcuts bound model action; reply tiers and token budgets match complexity; session memory is context, not fact.

Intelligent Information Solutions Center, Sharif University (IISC)

Tehran, Iran

AI Project Lead & Solution Architect

Jul 2019 – Sep 2025

- Led multidisciplinary delivery of AI products from research-style models into production, including **Mooshekaf** (natural-language access to databases, tabular data, and knowledge graphs) and an LLM-backed financial research chatbot.
- Directed a petroleum and commodity forecasting program (time-series + news deep learning; ~10% accuracy gain) and an RL portfolio allocator that reached ~90% of top-expert performance.
- Architected data/AI platforms: Kappa/Lambda lake for Tehran's Intelligent Traffic Management System, and enterprise data architecture for PED using TOGAF/ArchiMate.

Tali.ai (Intelligent Physician Assistant)

Toronto · Remote

NLP Scientist

Apr 2022 – Mar 2023

- Delivered a medical QA language model (F1 68%→79%); FAISS retrieval 3.9s→0.4s; MLOps on VertexAI + DVC cut deploy time from 3 days to 6 hours.

Ebuynow Co

Vancouver · Remote

NLP Specialist & Software Developer

Sep 2020 – Apr 2022

- Led aspect-based sentiment analysis on Amazon reviews and the surrounding data/IT architecture (Scrapy/Zyte, Kafka, AWS-hosted models, custom labeling workflow).

Port and Maritime Organization (PMO)

Tehran, Iran

Senior Consultant

Mar 2018 – Jun 2019

- Advised the CEO on Smart Port strategy: IoT and data-science integration across Iranian ports, plus data architecture guidance.

Municipality of Tehran, Real Estate Organization

Tehran, Iran

Head of IT Department

Sep 2011 – Sep 2017

- Promoted to Head of IT within 6 months; led Data Center and ERP programs and a property-acquisition data warehouse/ETL (Pentaho) across 22 municipal regions.

Education & PhD Research

Sharif University of Technology

2017 – Present

PhD Candidate, Software Engineering (GPA 4.00/4.00)

Supervisor: Prof. Jafar Habibi. **PRISM**: multi-hop QA over documents with **knowledge graphs**, **RAG**, and **plan-execute** LLM reasoning; provenance and entity grounding over scoped evidence.

Sharif University of Technology

2008 – 2010

MSc, Information Technology Engineering (GPA 3.81/4.00)

Thesis: *Relation Extraction from Text Datasets* (Supervisor: Prof. Hassan Abolhassani).

Teaching, Honors & Publications

- **Teaching (Sharif)**: Lecturer, Advanced Programming (2022–2023); TA for Advanced Software Architecture, Enterprise Architecture, Decision Support Systems, Data Mining, and Semantic Web.
- **Honors**: Rank 2 / ~12,000, National Master's Entrance Exam (2008); Rank 5 / ~1,500, National PhD Entrance Exam (2017).
- **Publications**: Relation extraction via sequential pattern mining (*CSI Conference*, 2013); DRL experts-advice aggregation for portfolio management (submitted, *Annals of Operations Research*, 2022); VAE arrhythmia detection with mmWave radar (submitted, *IEEE MTT*, 2022).